USAMA **ALI**

AI Engineer · ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building AI applications using Python, RAG pipelines, LLM APIs, embeddings, and machine learning. Experienced in FastAPI backend development, PostgreSQL, semantic search, and integrating AI technologies into practical applications.

TOOLS & TECHNOLOGY

- **Programming Languages:**
  Python
- **AI & Machine Learning:**
  Agents - RAG - LLMs - Langchain-LangGraph - Fine-Tuning - Transformer - NLP - ANNs - Supervised Unsupervised Learning - Hyperparameter Tuning - Model Evaluation - EDA - Speech-to-Text - Multimodal-AI - MLFlow - Docker - CI/CD - EC2,S3
- **Backend & Security:**
  FastAPI, REST APIs, SQLAlchemy, Alembic, JWT Authentication, Authorization
- **Databases:**
  PostgreSQL, SQLite
- **Vector Databases:**
  pgvector, ChromaDB, FAISS
- **Tools:**
  VSCode - Jupyter Notebook - Git - GitHub - MySQL - Ollama

EXPERIENCE

AI Engineer — Mindly (Startup) *February – March 2026*

- Built a knowledge-based AI system using RAG, embeddings, and LLM APIs for context-aware responses.
- Implemented document ingestion and semantic retrieval using Python.
- Website: mindly.one

GenAI Engineer Intern — CypherShield *July – November 2025*

- Contributed to GenAI application workflows by integrating and working with LLM APIs.
- Built automation workflows using N8N to streamline recurring development tasks.
- Collaborated with team members to communicate requirements and support GenAI development activities.

Other Projects

RepoGuide — AI-Powered Codebase Q&A

- Implemented repository ingestion, code-aware chunking, embeddings, semantic search, and reranking using Python, FastAPI, pgvector, Gemini, Jina AI, and LangChain, LangGraph.
- Implemented PostgreSQL persistence using SQLAlchemy and Alembic migrations for users, repositories, conversations, and messages.
- GitHub: RepoGuide

TutorDesk — AI Teaching Assistant

- Built an AI teaching assistant using RAG, LangChain,langGraph,Gemini, and structured LLM outputs for course-based question answering, quizzes, assignments, lesson plans, and explanations.
- Built a speech-to-text pipeline with browser audio capture, Gemini multimodal processing, validation, and RAG agent integration.
- Implemented course material RAG using PDF extraction, chunking, Gemini embeddings, PostgreSQL with PGVector, and per-user retrieval.
- Implemented an agent that routes requests between course-material RAG and web research using LangChain tools and Tavily.
- GitHub: TutorDesk

Movie Recommendation System

- Built a collaborative filtering movie recommendation system using Singular Value Decomposition (SVD).
- Implemented data preprocessing, model training, and recommendation generation using Python, Scikit-learn, and Surprise.
- GitHub: Movie Recommendation System

EDUCATION

Bachelor of Science in Computer Science *2022 – 2026*

University of Management and Technology, Lahore